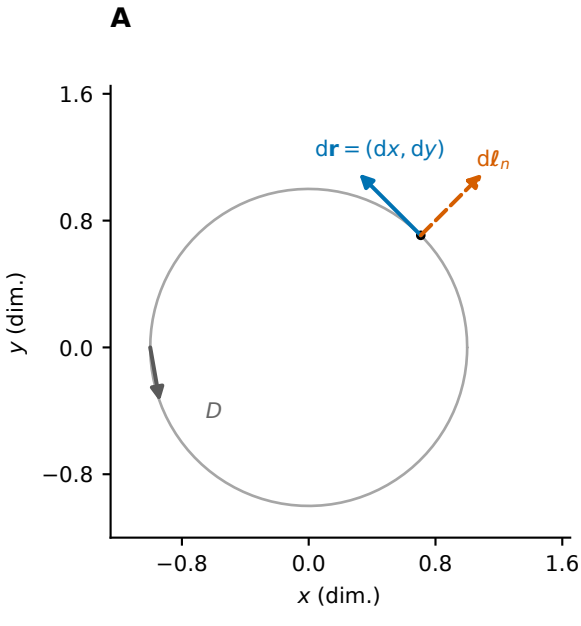
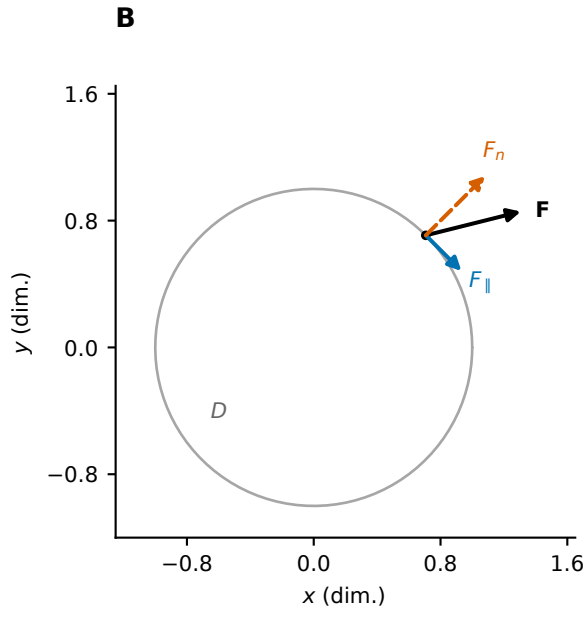


Positive tangent; outward normal is on its right



$\mathrm{d}\boldsymbol{\ell}_n = (\mathrm{d}y,\ -\mathrm{d}x)$; $\mathbf{F}\cdot\mathrm{d}\boldsymbol{\ell}_n = P\,\mathrm{d}y - Q\,\mathrm{d}x$